MINERAL AND ENERGY RESOURCES: WORLD AND INDIA

GEOLOGICAL CONTROL OF RESOURCE → MINERAL AND ENERGY CLASSIFICATION → WORLD MINERAL-ENERGY BELT RAIL → EXTRACTION TO INDUSTRY CHAIN → OCCURRENCE TO USABLE RESOURCE → FOUR-ELEMENT PETROLEUM SYSTEM

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ● SUBORDINATE EXTRA ★ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g4 and all prior artifacts unchanged

00 GEOLOGICAL CONTROL OF RESOURCE DISTRIBUTION

STAGE 00 · GEOLOGICAL CONTROL OF RESOURCE DISTRIBUTION · GEOLOGICAL HISTORY · ROCK STRUCTURE, BASINS, TECTONICS, SURFACE WEATHERING · IGNEOUS-METAMORPHIC TERRAIN · METALLIC ORES · SEDIMENTARY BASIN · COAL, PETROLEUM, NATURAL GAS · TROPICAL LATERITISATION

GEOLOGICAL HISTORY — rock structure, basins, tectonics, surface weathering → SHIELD / IGNEOUS-METAMORPHIC TERRAIN — metallic ores → SEDIMENTARY BASIN — coal, petroleum, natural gas → TROPICAL LATERITISATION — bauxite on suitable plateaux and hilltops → TRAP — distribution is a deduction from Earth history, not a country list

CORE CLAIM
- GEOLOGICAL HISTORY → rock structure, basins, tectonics, surface weathering
- SHIELD / IGNEOUS-METAMORPHIC TERRAIN → metallic ores
- SEDIMENTARY BASIN → coal, petroleum, natural gas

MECHANISM
- TROPICAL LATERITISATION → bauxite on suitable plateaux and hilltops
- TRAP → distribution is a deduction from Earth history, not a country list
- MUST REMEMBER: Resource geography begins with geological occurrence, reserve/resource classification, grade, extraction technology, transport, processing, markets and externalities

CONSEQUENCE / LIMIT
- world belts and India's Gondwana coal, offshore/onshore hydrocarbons.
- Odisha–Jharkhand iron-manganese-bauxite, Rajasthan non-ferrous minerals and nuclear/renewable endowments require map logic.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → distribution is a deduction from Earth history, not a country list.

01 MINERAL AND ENERGY CLASSIFICATION LADDER

STAGE 01 · MINERAL AND ENERGY CLASSIFICATION LADDER · FERROUS METALLIC · IRON ORE, MANGANESE, CHROMITE; STEEL-MAKING BASE · NON-FERROUS METALLIC · COPPER, BAUXITE, LEAD, ZINC · NON-METALLIC INDUSTRIAL · MICA, LIMESTONE, GYPSUM, PHOSPHATES · MINERAL FUELS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
FERROUS METALLIC → iron ore, manganese, chromite; steel-making base	NON-METALLIC INDUSTRIAL → mica, limestone, gypsum, phosphates	ENERGY PAIRS → conventional vs non-conventional; exhaustible stock vs renewable flow
NON-FERROUS METALLIC → copper, bauxite, lead, zinc	MINERAL FUELS → coal, petroleum, natural gas, uranium	FERROUS METALLIC → iron ore, manganese, chromite; steel-making base

CORE CLAIM
- FERROUS METALLIC → iron ore, manganese, chromite; steel-making base
- NON-FERROUS METALLIC → copper, bauxite, lead, zinc

MECHANISM
- NON-METALLIC INDUSTRIAL → mica, limestone, gypsum, phosphates
- MINERAL FUELS → coal, petroleum, natural gas, uranium

CONSEQUENCE / LIMIT
- ENERGY PAIRS → conventional vs non-conventional; exhaustible stock vs renewable flow
- FERROUS METALLIC → iron ore, manganese, chromite; steel-making base

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ENERGY PAIRS → conventional vs non-conventional; exhaustible stock vs renewable flow.

02 WORLD MINERAL-ENERGY BELT RAIL

STAGE 02 · WORLD MINERAL-ENERGY BELT RAIL · PERSIAN GULF · CLASSIC SEDIMENTARY HYDROCARBON PROVINCE · INTERIOR USA · COAL, OIL, GAS WITH IRON ASSOCIATIONS · HISTORIC COAL-IRON HEAVY-INDUSTRY CORE · CHINESE INTERIOR · CONTINENTAL-SCALE FUEL FRONTIERS

PERSIAN GULF — classic sedimentary hydrocarbon province

APPALACHIAN / INTERIOR USA — coal, oil, gas with iron associations → RUHR-LORRAINE — historic coal-iron heavy-industry core → RUSSIAN-SIBERIAN / CHINESE INTERIOR — continental-scale fuel frontiers

CORE CLAIM
- PERSIAN GULF → classic sedimentary hydrocarbon province
- APPALACHIAN / INTERIOR USA → coal, oil, gas with iron associations

MECHANISM
- RUHR-LORRAINE → historic coal-iron heavy-industry core
- RUSSIAN-SIBERIAN / CHINESE INTERIOR → continental-scale fuel frontiers

CONSEQUENCE / LIMIT
- COPPERBELT, AUSTRALIAN SHIELD, ANDES → shield and tectonic metal provinces
- PERSIAN GULF → classic sedimentary hydrocarbon province

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COPPERBELT, AUSTRALIAN SHIELD, ANDES → shield and tectonic metal provinces.

03 EXTRACTION TO INDUSTRY CHAIN

STAGE 03 · EXTRACTION TO INDUSTRY CHAIN · ORE OR FUEL RAISED AT THE DEPOSIT · RAIL, SLURRY PIPELINE, PORT OR INLAND WATERWAY EVACUATION · BENEFICIATION, SMELTING, REFINING, PETROCHEMICAL CONVERSION · POWER + MARKET · DISPATCHABLE ENERGY, CAPITAL AND DEMAND COMPLETE THE CLUSTER · A PRODUCING REGION IS NOT AUTOMATICALLY AN INDUSTRIAL REGION

EXTRACTION — ore or fuel raised at the deposit → TRANSPORT — rail, slurry pipeline, port or inland waterway evacuation → PROCESSING — beneficiation, smelting, refining, petrochemical conversion → POWER + MARKET — dispatchable energy, capital and demand complete the cluster → RULE — a producing region is not automatically an industrial region

CORE CLAIM
- EXTRACTION → ore or fuel raised at the deposit
- TRANSPORT → rail, slurry pipeline, port or inland waterway evacuation

MECHANISM
- PROCESSING → beneficiation, smelting, refining, petrochemical conversion
- POWER + MARKET → dispatchable energy, capital and demand complete the cluster

CONSEQUENCE / LIMIT
- RULE → a producing region is not automatically an industrial region
- EXTRACTION → ore or fuel raised at the deposit

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → a producing region is not automatically an industrial region.

04 OCCURRENCE TO USABLE RESOURCE FILTER

STAGE 04 · OCCURRENCE TO USABLE RESOURCE FILTER · OCCURRENCE EXISTS · GEOLOGY HAS DONE ITS PART · IF GRADE AND DEPTH ARE WORKABLE · CONTINUE TO THE ECONOMIC TEST · IF TRANSPORT DISTANCE AND TECHNOLOGY PERMIT · CONTINUE TO THE PRICE TEST · IF PRICE SUSTAINS THE COST

OCCURRENCE EXISTS — geology has done its part

If grade and depth are workable — continue to the economic test → If transport distance and technology permit — continue to the price test → If price sustains the cost — occurrence becomes a usable resource

CORE CLAIM
- OCCURRENCE EXISTS → geology has done its part
- IF grade and depth are workable → continue to the economic test

MECHANISM
- IF transport distance and technology permit → continue to the price test
- IF price sustains the cost → occurrence becomes a usable resource

CONSEQUENCE / LIMIT
- ELSE → the deposit stays a geological fact with no economic geography
- OCCURRENCE EXISTS → geology has done its part

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ELSE → the deposit stays a geological fact with no economic geography.

05 FOUR-ELEMENT PETROLEUM SYSTEM

STAGE 05 · FOUR-ELEMENT PETROLEUM SYSTEM · SOURCE ROCK · ORGANIC-RICH FINE SEDIMENT BURIED DEEPLY AND LONG ENOUGH · RESERVOIR ROCK · POROUS PERMEABLE SANDSTONE OR FRACTURED CARBONATE · CAP ROCK · SHALE, EVAPORITE OR TIGHT LIMESTONE PREVENTING ESCAPE · ANTICLINE, FAULT BLOCK, SALT DOME, REEF OR STRATIGRAPHIC PINCH-OUT

SOURCE ROCK — organic-rich fine sediment buried deeply and long enough → RESERVOIR ROCK — porous permeable sandstone or fractured carbonate → SEAL / CAP ROCK — shale, evaporite or tight limestone preventing escape → TRAP — anticline, fault block, salt dome, reef or stratigraphic pinch-out → VERDICT — sedimentary basin is necessary but never sufficient

CORE CLAIM
- SOURCE ROCK → organic-rich fine sediment buried deeply and long enough
- RESERVOIR ROCK → porous permeable sandstone or fractured carbonate

MECHANISM
- SEAL / CAP ROCK → shale, evaporite or tight limestone preventing escape
- TRAP → anticline, fault block, salt dome, reef or stratigraphic pinch-out

CONSEQUENCE / LIMIT
- VERDICT → sedimentary basin is necessary but never sufficient
- SOURCE ROCK → organic-rich fine sediment buried deeply and long enough

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → sedimentary basin is necessary but never sufficient.

06 WHY PASSIVE MARGINS HOLD OFFSHORE OIL

STAGE 06 · WHY PASSIVE MARGINS HOLD OFFSHORE OIL · SHELF AND SLOPE · THE WORLD'S GREAT SEDIMENT SINKS RECEIVE RIVER-BORNE ORGANIC LOAD · OXYGEN-POOR BOTTOM WATER · MARINE SETTINGS PRESERVE ORGANIC MATTER FROM DECAY · PASSIVE MARGIN SUBSIDENCE · THICK, TECTONICALLY QUIET SEDIMENT PRISMS ACCUMULATE · DELTAS AND SALT TECTONICS

SHELF AND SLOPE — the world's great sediment sinks receive river-borne organic load → OXYGEN-POOR BOTTOM WATER — marine settings preserve organic matter from decay → PASSIVE MARGIN SUBSIDENCE — thick, tectonically quiet sediment prisms accumulate → DELTAS AND SALT TECTONICS — pair source with reservoir and generate trap families → PRESERVATION EDGE — uplift and erosion have breached many onshore accumulations

CORE CLAIM
- SHELF AND SLOPE → the world's great sediment sinks receive river-borne organic load
- OXYGEN-POOR BOTTOM WATER → marine settings preserve organic matter from decay

MECHANISM
- PASSIVE MARGIN SUBSIDENCE → thick, tectonically quiet sediment prisms accumulate
- DELTAS AND SALT TECTONICS → pair source with reservoir and generate trap families

CONSEQUENCE / LIMIT
- PRESERVATION EDGE → uplift and erosion have breached many onshore accumulations
- SHELF AND SLOPE → the world's great sediment sinks receive river-borne organic load

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DELTAS AND SALT TECTONICS → pair source with reservoir and generate trap families.

07 OFFSHORE VERSUS ONSHORE COMPARISON

STAGE 07 · OFFSHORE VERSUS ONSHORE COMPARISON · ONSHORE INTRACRATONIC AND RIFT BASINS VS SHELF, SLOPE AND · COST AND TECHNOLOGY · INCREMENTAL LAND DRILLING VS LUMPY HIGH-COST MARINE SYSTEMS · SMALL ONSHORE FIELDS CAN BE VIABLE; OFFSHORE NEEDS LARGE · LAND ACQUISITION ONSHORE VS FISHERIES, SHIPPING AND SPILL RISK · ORDINARY NATIONAL LAW VS MARITIME ZONES AND CONTESTED BOUNDARIES

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
GEOLOGY → onshore intracratonic and rift basins vs shelf, slope and deepwater margin	THRESHOLD → small onshore fields can be viable; offshore needs large fields	JURISDICTION → ordinary national law vs maritime zones and contested boundaries
COST AND TECHNOLOGY → incremental land drilling vs lumpy high-cost marine systems	FOOTPRINT → land acquisition onshore vs fisheries, shipping and spill risk offshore	GEOLOGY → onshore intracratonic and rift basins vs shelf, slope and deepwater margin

CORE CLAIM
- GEOLOGY → onshore intracratonic and rift basins vs shelf, slope and deepwater margin
- COST AND TECHNOLOGY → incremental land drilling vs lumpy high-cost marine systems

MECHANISM
- THRESHOLD → small onshore fields can be viable; offshore needs large fields
- FOOTPRINT → land acquisition onshore vs fisheries, shipping and spill risk offshore

CONSEQUENCE / LIMIT
- JURISDICTION → ordinary national law vs maritime zones and contested boundaries
- GEOLOGY → onshore intracratonic and rift basins vs shelf, slope and deepwater margin

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: JURISDICTION → ordinary national law vs maritime zones and contested boundaries.

08 INDIA PETROLEUM PROVINCE MAP HOOKS

STAGE 08 · INDIA PETROLEUM PROVINCE MAP HOOKS · ASSAM-ARAKAN BASIN AND THE CAMBAY BASIN OF WESTERN INDIA · WESTERN OFFSHORE · MAJOR CONTINENTAL-MARGIN PROVINCE, CARBONATE RESERVOIR CHARACTER · EAST-COAST OFFSHORE · KRISHNA-GODAVARI AND CAUVERY DELTAIC MARGINS, GAS-PRONE · COASTAL CONSEQUENCE · LANDFALL TERMINALS PULL REFINING AND PETROCHEMICALS TO THE COAST

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
ONSHORE → Assam-Arakan basin and the Cambay basin of western India	EAST-COAST OFFSHORE → Krishna-Godavari and Cauvery deltaic margins, gas-prone	CAUTION → name provinces and reservoir character, never remembered output figures
WESTERN OFFSHORE → major continental-margin province, carbonate reservoir character	COASTAL CONSEQUENCE → landfall terminals pull refining and petrochemicals to the coast	ONSHORE → Assam-Arakan basin and the Cambay basin of western India

CORE CLAIM
- ONSHORE → Assam-Arakan basin and the Cambay basin of western India
- WESTERN OFFSHORE → major continental-margin province, carbonate reservoir character

MECHANISM
- EAST-COAST OFFSHORE → Krishna-Godavari and Cauvery deltaic margins, gas-prone
- COASTAL CONSEQUENCE → landfall terminals pull refining and petrochemicals to the coast

CONSEQUENCE / LIMIT
- CAUTION → name provinces and reservoir character, never remembered output figures
- ONSHORE → Assam-Arakan basin and the Cambay basin of western India

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CAUTION → name provinces and reservoir character, never remembered output figures.

09 CRITICAL-MINERAL CHOKEPOINT FORK

STAGE 09 · CRITICAL-MINERAL CHOKEPOINT FORK · TRANSITION DEMAND RISES · LITHIUM, COBALT, NICKEL, GRAPHITE, COPPER, RARE EARTHS · IF THE QUESTION IS EXTRACTION · SUPPLY IS GEOLOGICALLY NARROW BUT VISIBLE · IF THE QUESTION IS PROCESSING · CONCENTRATION IS TIGHTER AND OFTEN IN OTHER COUNTRIES · THE BINDING CHOKEPOINT IS MIDSTREAM, NOT AT THE MINE

TRANSITION DEMAND RISES — lithium, cobalt, nickel, graphite, copper, rare earths

If the question is extraction — supply is geologically narrow but visible → If the question is processing — concentration is tighter and often in other countries → THEREFORE — the binding chokepoint is midstream, not at the mine

CORE CLAIM
- TRANSITION DEMAND RISES → lithium, cobalt, nickel, graphite, copper, rare earths
- IF the question is extraction → supply is geologically narrow but visible
- IF the question is processing → concentration is tighter and often in other countries

MECHANISM
- THEREFORE → the binding chokepoint is midstream, not at the mine
- RESPONSE → recycling, substitution, diversification and strategic stockpiling
- CLOSE DISTINCTION: Resource differs from reserve, occurrence from economically recoverable deposit, thermal from metallurgical coal, conventional from non-conventional

CONSEQUENCE / LIMIT
- criticality combines economic importance and supply risk rather than geological scarcity alone.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: THEREFORE → the binding chokepoint is midstream, not at the mine.

10 INDIA MINERAL BELT AND ENERGY CORRIDOR RAIL

STAGE 10 · INDIA MINERAL BELT AND ENERGY CORRIDOR RAIL · CHOTA NAGPUR COMPLEX · IRON ORE, COAL, MICA, BAUXITE, MANGANESE HEARTLAND · BELT HOOKS · BAILADILA IRON, JHARIA COKING COAL, KHETRI COPPER, ZAWAR LEAD-ZINC · GONDWANA COAL · DAMODAR, MAHANADI, SON AND GODAVARI VALLEY BASINS · SOLAR CORRIDOR

CHOTA NAGPUR COMPLEX — iron ore, coal, mica, bauxite, manganese heartland

BELT HOOKS — Bailadila iron, Jharia coking coal, Khetri copper, Zawar lead-zinc → GONDWANA COAL — Damodar, Mahanadi, Son and Godavari valley basins → SOLAR CORRIDOR — Rajasthan and Gujarat dry high-insolation belts

CORE CLAIM
- CHOTA NAGPUR COMPLEX → iron ore, coal, mica, bauxite, manganese heartland
- BELT HOOKS → Bailadila iron, Jharia coking coal, Khetri copper, Zawar lead-zinc
- GONDWANA COAL → Damodar, Mahanadi, Son and Godavari valley basins

MECHANISM
- SOLAR CORRIDOR → Rajasthan and Gujarat dry high-insolation belts
- WIND CORRIDOR → Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes
- EVIDENCE LIMIT: Jharia/Raniganj/Talcher/Korba, Mumbai High, Assam, Krishna-Godavari,

CONSEQUENCE / LIMIT
- Bailadila, Keonjhar, Khetri, Jaduguda, monazite coasts; solar Rajasthan/Gujarat and wind
- Tamil Nadu/Gujarat anchor India
- Ministry, IBM, CEA and international data need year/status labels, while just-transition and import dependence qualify security claims.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Tamil Nadu/Gujarat anchor India; Ministry, IBM, CEA and international data need year/status labels, while just-transition and import dependence qualify security claims.

11 MINERAL AND ENERGY ANSWER SPINE

STAGE 11 · MINERAL AND ENERGY ANSWER SPINE · WHICH CLASS OF RESOURCE AND WHICH MEASURE IS BEING · SHIELD, BASIN OR WEATHERING CONTROL BEFORE NAMING ANY PLACE · WORLD BELT, THEN THE INDIAN BELT OR PETROLEUM PROVINCE · OCCURRENCE VERSUS USABLE RESOURCE; CAPACITY VERSUS GENERATION · MIDSTREAM CHOKEPOINT, INSTITUTIONAL CONVERSION AND THE DATED MISSION ANCHOR

DEFINE — which class of resource and which measure is being asked about → DEDUCE — shield, basin or weathering control before naming any place → LOCATE — world belt, then the Indian belt or petroleum province → QUALIFY — occurrence versus usable resource; capacity versus generation → CLOSE — midstream chokepoint, institutional conversion and the dated mission anchor

CORE CLAIM
- DEFINE → which class of resource and which measure is being asked about
- DEDUCE → shield, basin or weathering control before naming any place

MECHANISM
- LOCATE → world belt, then the Indian belt or petroleum province
- QUALIFY → occurrence versus usable resource; capacity versus generation

CONSEQUENCE / LIMIT
- CLOSE → midstream chokepoint, institutional conversion and the dated mission anchor
- DEFINE → which class of resource and which measure is being asked about

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE → midstream chokepoint, institutional conversion and the dated mission anchor.

E SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

EXTRA · OPTIONAL ADVANCED ENRICHMENT · INDIA IS STRONG IN SEVERAL BULK MINERALS BUT STILL · DEPENDENCE IN MANY CRITICAL MINERALS NEEDED FOR BATTERIES, ELECTRONICS · BULK-MINERAL STRENGTH DOES NOT AUTOMATICALLY TRANSLATE INTO HIGH-END PROCESSING · STRATEGIC AUTONOMY NOW REQUIRES CONTROL OVER EXPLORATION, BENEFICIATION, PROCESSING · EAST-CENTRAL MINERAL ABUNDANCE ALSO CREATES A LOGISTICS QUESTION · RAIL EVACUATION · SLURRY PIPELINES, PORT LINKAGES AND POWER AVAILABILITY DECIDE ACTUAL

OPTIONAL SOURCE DEPTH
- CAUTION: India is strong in several bulk minerals but still faces external
- dependence in many critical minerals needed for batteries, electronics and clean energy.
- CAUTION: Bulk-mineral strength does not automatically translate into high-end processing strength.
- CAUTION: Strategic autonomy now requires control over exploration, beneficiation, processing and recycling , not only extraction.

ADVANCED DEBATE
- CAUTION: East-central mineral abundance also creates a logistics question; rail evacuation,
- slurry pipelines, port linkages and power availability decide actual industrial value.
- Chota Nagpur Plateau is India's classic mineral heartland.
- Coalfields strongly influenced the location of thermal power and heavy industry.

LEGACY / NUANCE
- Rajasthan is especially important for several non-ferrous and industrial minerals.
- Hydrocarbons are associated with sedimentary basins and offshore areas, not with shield-core metallic belts.
- Solar and wind geography do not mirror coal geography.
- NOT: India's mineral core lies in the Indo-Gangetic plain - It is mainly in the peninsular shield and associated belts.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels